

Avatar Motion Generation Pipeline for the Metaverse via Synthesis of Generative Models of Text and Video

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Figure 1: Left: An AI avatar in the metaverse in the default pose before any motion is applied. Center: Video frames generated from the user's text instructions by the video generation model. Right: The AI avatar is animated using pose data extracted from the generated video via the pose recognition model.

ABSTRACT

Efforts to integrate AI avatars into the metaverse to enhance interactivity have progressed in both research and commercial domains. AI avatars in the metaverse are expected to exhibit not only verbal responses but also avatar motions, such as non-verbal gestures, to enable seamless communication with users. Large Language Models (LLMs) are known for their advanced text processing capabilities, such as user input, avatar actions, and even entire virtual environments as text, making them a promising approach for planning avatar motions. However, generating the avatar motions solely from the textual information often requires extensive training data whereas the configuration is very challenging, with results that often lack diversity and fail to match user expectations. On the other hand, AI technologies for generating videos have progressed to the point where they can depict diverse and natural human movements based on prompts. Therefore, this paper introduces a novel pipeline, TVMP, that synthesizes LLMs with advanced text processing capabilities and video generation models with the ability to generate videos containing a variety of motions. The pipeline first generates videos from text input, then estimates the motions from the generated videos, and lastly exports the estimated motion data into the avatars in the metaverse. Feedback on the TVMP prototype suggests further refinement is needed, such as speed control, display of the progress, and direct edition for contextual relevance and usability enhancements. The proposed method enables AI avatars to perform highly adaptive and diverse movements to fulfill user expectations and contributes to developing a more immersive metaverse.

Index Terms: Human-centered computing—Visualization—Visualization techniques—Treemaps; Human-centered computing—Visualization—Visualization design and evaluation methods

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1 INTRODUCTION

Efforts to integrate AI into avatars (i.e., virtual representations of bodies in digital environments) have advanced in both academic and commercial domains [1, 5, 12]. These AI avatars are designed to exhibit not only verbal responses but also natural motions, facilitating more intuitive communication with users [1, 5]. To achieve this, using Large Language Models (LLMs) with sophisticated text processing capabilities has been explored to simulate aspects of human cognition [13, 16]. Furthermore, generating non-verbal motions for AI avatars using Text-to-Motion models has been investigated, where descriptive text is translated into motion sequences [1, 5].

However, the traditional approach lacks the diversity and complexity of motions for AI avatars. Since Text-to-Motion models rely on a limited dataset of text-motion pairs, generating motions from unknown textual inputs of motion fails most of the time. Furthermore, models generating diverse motions [2, 3, 9] are often large-scale. Such models require significant computational resources and complex setups, hindering usability for many users.

On the other hand, video generation models which generate videos from images (Image-to-Video Models) or texts (Text-to-Video Models) are generally trained from larger datasets than Text-to-Motion, executable with smaller computational resources and setups with ease. Additionally, video generation models are already commercially available as APIs (e.g., Sora [10], Runway [14], KLING [7] etc.), which provides high usability. Nevertheless, the method to derive a moving AI avatar in metaverse from a user input is yet to be known considering the data processing complexity.

This study proposes a novel pipeline, Text-to-Video-to-Motion pipeline (TVMP), which integrates motions extracted from video generation models based on user instructions and LLM instructions, into AI avatars in the metaverse. TVMP begins by generating keyframe images from user text input, synthesizing the generated images with the original user text to create videos. Subsequently, it estimates motion data from the generated video and converts this data to align with the avatar coordinates of the target metaverse platform, which are then exported and applied to the avatar.

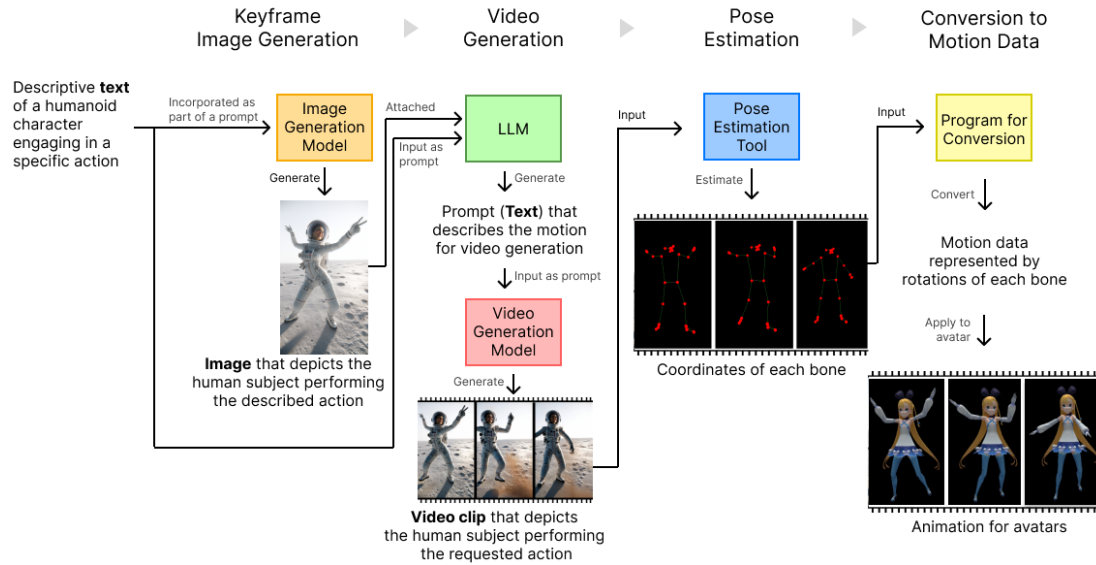


Figure 2: System architecture of the proposed pipeline

2 SYSTEM ARCHITECTURE

The pipeline consists of the following three stages in order: keyframe image generation from text-based user input, video generation from the keyframe image, pose estimation from the video, and conversion to motion data (Figure 2).

The entire pipeline was implemented as an API-based service structure to allow external applications like metaverse platforms to send a single text description and receive detailed motion data as a response. Furthermore, by separating each stage of the pipeline into distinct modules, our system can be extended or modified (e.g., to accommodate different pose-estimation libraries or image/video generation APIs) without fundamentally altering the workflow.

2.1 Keyframe Image Generation

Upon receiving a descriptive text of a humanoid character engaging in a specific action (e.g., “an astronaut dancing on the moon”), the pipeline is initiated from the image generation stage.

To avoid impacting the quality of the later pose estimation process, the generated image should be ensured with a plain background and capturing the full body of the human subject. Following this, the system first incorporates the input description into a predefined prompt template. The prompt template shall be composed of ratio size of image, light condition, number of people which must be specified as one person, posture and visibility of the body, clothing, camera position and field of view, and a plain background. The example of predefined prompt template configured for OpenAI image generation API appears below:

A photorealistic image of {input description} at daytime. vertical portrait orientation, aspect 9:16. single subject, standing upright, wearing tight clothing. Clearly visible head, face, torso, two arms, and two legs. Camera stays far from the subject, capturing the full body from head to feet. Blank background. Remove surrounding people and objects.

The formatted prompt is then passed to an image generation model to produce a single keyframe image, which serves as the representation frame image of the motion. This image serves as the reference frame for the subsequent video generation step, ensuring that a video is generated that matches the context inferred from the text and image.

2.2 Video Generation

Once the keyframe image is obtained, it is passed to an LLM along with a predefined prompt. The example of predefined prompt template to generate a video generation prompt configured for the OpenAI API appears below:

Understand the article from this link about how to create prompts for Runway video generation: <https://help.runwayml.com/hc/en-us/articles/30586818553107-Gen-3-Alpha-Prompting-Guide> Then, generate a short (less than 500 characters) text prompt for Runway video generation of a human engaging in a specific action: "{userInput}". The attached image will be used as a keyframe of the video. The text prompt should include the following requirements for the video:

- Start from "Camera locked-off, static, still, with a fixed perspective and distance from the character, capturing its full body:
- Description to reflect the character's typical characteristics on its motions.
- Add distinct variance in the character's motions between keyframes, such as changes in poses, while keeping the actions consistent with the instruction.

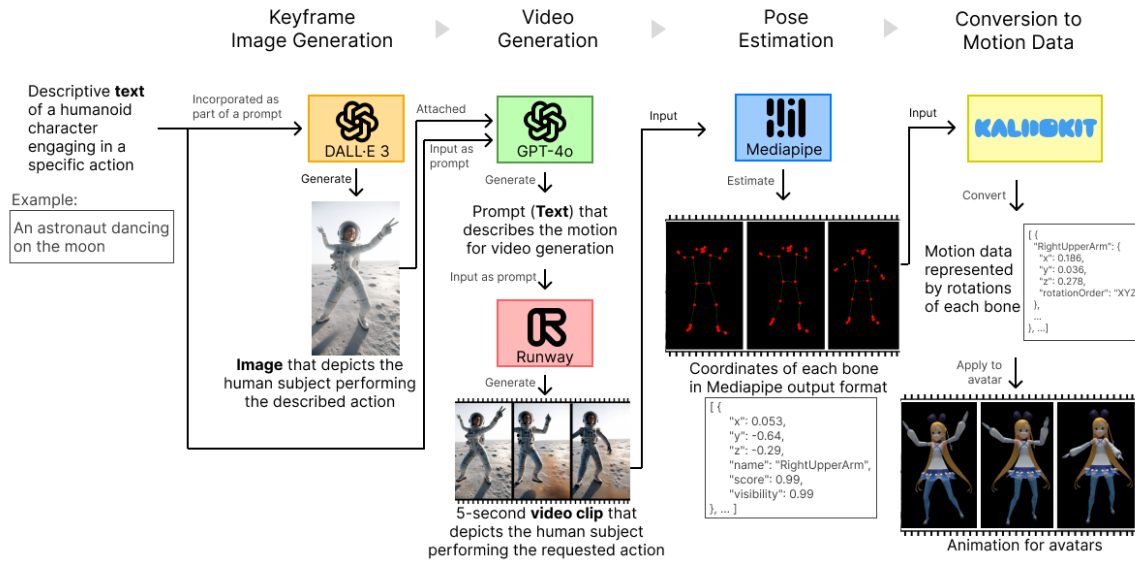


Figure 3: Prototype of the proposed pipeline

- Maintain high visual coherence and smooth transitions.

The LLM generates a refined prompt that guides the video generation process, ensuring a fixed camera perspective, consistent context, and no scene transitions. These considerations are essential for preventing unexpected results on the later pose estimation process. An example of generated prompt configured for video generation API appears below:

Camera locked-off, static, still, with a fixed perspective and distance from the character, capturing its full body: The character dances with dynamic flair, transitioning smoothly between powerful, angular poses and fluid, graceful gestures. Displays high energy with rhythmic movements, subtly shifting weight from one foot to another. Consistent focus on fluidity and expression, maintaining a seamless flow between keyframes while capturing the essence of an idol's captivating stage presence.

The generated image and prompt are then passed to a video generation model to produce a video clip that has a sufficient length to depict a human performing the user-requested motion. This video clip serves as an input for the pose estimation process.

2.3 Pose Estimation

Once the video clip is generated, each frame undergoes a pose estimation process, which identifies the human subject in each video frame and outputs 3D coordinates of the estimated joint positions. These 3D coordinates are then aggregated across all frames in the video.

2.4 Conversion to Motion Data

Rather than returning the raw joint positions, each joint's coordinates are converted into corresponding quaternions, yielding a detailed, ready-to-use motion data that represents the user-requested action.

The motion is finally included in the pipeline's API response and returned to the user.

3 PROTOTYPE

A prototype of the proposed pipeline (Figure 3) was developed using Node.js¹ with Express.js² to host the pipeline's API. In the keyframe image generation stage, the OpenAI image generation API was integrated to utilize the DALL-E 3 model³. During the video generation stage, OpenAI's GPT-4o model, accessed via OpenAI's assistant API⁴ was employed as the LLM, and a 5-second video was generated using Runway via their API⁵. For pose estimation, the system adopted MediaPipe BlazePose⁶ employed in conjunction with the TensorFlow.js pose detection API⁷. Finally, the coordinates of estimated poses into motion data utilized Kalidokit⁸, a kinematics solver library capable of transforming joint coordinates estimated by Mediapipe into quaternions.

Additionally, a custom adapter was developed to apply the API-exported motion data to VRM, a widely-used de facto standard for 3D avatars, ensuring compatibility with various metaverse platforms such as Hubs (a WebVR-based metaverse platform, successor of Mozilla Hubs)⁹ and Cluster¹⁰. Once the motion data is adapted to VRM, users can see the AI avatar performing the requested motions in real-time. Though VRM is the initial target format, the system is designed to be extensible; future improvements will include support for additional 3D file formats to broaden compatibility with various metaverse or for other purposes.

A proof-of-concept integration is implemented on chutvrc [4] which is a forked metaverse system from the Mozilla Hubs, devel-

¹<https://nodejs.org>

²<https://expressjs.com>

³<https://platform.openai.com/docs/guides/images>

⁴<https://platform.openai.com/docs/assistants/overview>

⁵<https://docs.dev.runwayml.com/api-details/sdks>

⁶<https://ai.google.dev/edge/mediapipe/solutions/guide>

⁷<https://github.com/tensorflow/tfjs-models/tree/master/pose-detection>

⁸<https://github.com/yeemachine/kalidokit>

⁹<https://hubsfoundation.org>

¹⁰<https://cluster.mu>

oped with functional enhancement on avatars and avatar communications. The proof-of-concept system demonstrates the pipeline's API calling and the application of the generated motion on an avatar within the metaverse.

4 TVMP PROTOTYPE TESTING AND FEEDBACK

The developed pipeline was tested on a system level using the proof-of-concept system on chutvrc. Due to the nature of the system to rely on user input to start the system, a real human tester was asked to try the system and provide feedback. The feedback focused on two key aspects: the quality of the generated motions and the system's usability.

The quality of the generated motions generally adhered to the prompt instructions, feedback indicated a discrepancy concerning their appropriateness to the avatar's persona and contextual relevance. This suggests that further advancements in both image and video generation components are essential to achieve contextually relevant motions. Besides, this may also be improved by not only refining the prompt text to include instructions to avoid discrepancy, but also developing and evaluating a stronger mechanism to better understand the AI avatar's attributes, situation, and surrounding environment.

The feedback for system usability were functional proposal of lacking features to control the motion in detail, which was out of the scope from the system level proof-of-concept testing. The proposed features were functions to control the speed of motion, to display the progress of generation, and to edit the prompt templates directly. Considering the importance of system usability, features to control the motion in detail shall provide results more closely aligned with the intention of the user input. Therefore, implementing such features is to be considered for higher proactive user engagements with TVMP, ultimately improving system usability.

5 LIMITATION AND FUTURE WORK

TVMP is a system that facilitates the generation of AI avatar motions by synthesizing image and video generation models. However, the pipeline's performance can be affected by issues in the libraries it depends on. For example, occasional terminations occur due to issues with OpenAI's image generation, such as content policy violation errors and rotated orientations in the output images. Furthermore, issues remain regarding the time required for generation (1 minute per prompt) and unnatural vibrations in the generated motions of intense movements such as jumping. In future work, filters like low-pass filter could be applied to the generated motions to eliminate unnatural vibrations.

The proposed prototype and proof-of-concept system could be improved and enhanced based on the feedback for reliable testing and future user studies. Although issues concerning generation time and unnatural movements can be addressed through advancements in hardware and the development of video generation by integrating 3D motion information [6, 8], future work on enabling seamless synthesis of AI avatar motions remains necessary.

Leveraging LLMs and video generation technologies like DALL-E 3 and Runway to create diverse AI avatar motions for metaverse environments, ultimately enhancing user experiences. Moreover, TVMP seeks support for not only virtual reality (VR) research but also contribution to the revitalization of real-time streaming [15] and communication in VR SNS [11] fostering the development of the metaverse.

6 CONCLUSION

This study proposed a novel pipeline, TVMP, which integrates motion from video generation models based on user instructions and LLM instructions, into AI avatars in the metaverse. Addressing the limitations of traditional Text-to-Motion models reliant on limited datasets and computationally intensive setups. The TVMP prototype was developed as an API-based service generating motion data from

user text input. The API-based implementation allows TVMP to evolve in line with advancements in video generation models, ensuring the integration of cutting-edge techniques. Feedback from prototype testing suggests that future work includes refining the image and video generation processes and incorporating user controls, such as motion speed adjustment and prompt editing functionalities. In conclusion, TVMP contributes to accelerating the metaverse in academic and commercial domains by providing a pipeline system that derives moving AI avatars from user input, thus facilitating the possibility for everyone wanting to create expressive AI avatars.

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